

Detecting Drug Interactions with Machine Learning

Progress Report — v6

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Project Overview

Drug-drug interactions (DDIs) are a leading cause of adverse drug events and a critical challenge in pharmacovigilance, particularly in polypharmacy settings. This project explores machine learning-based DDI prediction, building on the DDI-LLM framework [1] which combines LLM embeddings with graph convolutional networks (GCNs) for link prediction on benchmark datasets.

The central research question is: *can the performance of the LLM + GCN framework on OGB-ddi be improved by (1) substituting biomedical-specific LLM embeddings (PubMedBERT, ChemBERTa) for general-purpose ones, and (2) replacing random negative sampling with clinically verified non-interacting pairs from CRESCENDDI — and if so, by how much compared to the state-of-the-art result of $AUC = 0.900$ reported by LLM-DDI [2]?*

Related Work

Following the taxonomy of Qiu et al. [3], computational DDI detection methods fall into three categories: literature-based extraction (NLP on biomedical text), machine learning-based prediction (the focus of this project), and pharmacovigilance-based data mining. Within machine learning-based prediction, the most relevant prior works are:

DeepDDI [5] (Ryu et al., 2018) uses structural similarity profiles as input to a deep neural network. On DrugBank it reports AUC-ROC of 0.926.

KGNN [4] (Lin et al., 2020) proposes a knowledge graph neural network that captures topological neighbourhood information. On DrugBank (V5.1.4) it achieves AUC-ROC of 0.991. On OGB-biokg it achieves ACC of 0.695 and AUC of 0.820.

LLM-DDI [2] (Li et al., 2026) integrates LLM-generated molecular embeddings (text-embedding-ada-002, dimension 1536) into a GNN-based framework evaluated on OGB datasets using 5-fold cross-validation. Key results from Table II (p.778):

Method	OGB-biokg				OGB-ddi			
	ACC	F1	AUC	AUPR	ACC	F1	AUC	AUPR
RF	0.910	0.911	0.972	0.973	0.730	0.733	0.804	0.787
KGNN	0.695	0.579	0.820	0.814	0.511	0.549	0.515	0.556
KG2ECapsule	0.937	0.938	0.982	0.971	0.735	0.752	0.794	0.744
LLM-DDI	0.964	0.964	0.994	0.994	0.822	0.824	0.900	0.883

Datasets

BioSnap and DrugBank

Dataset	Drugs	Interactions	Sparsity	After filtering
BioSnap	1,514	48,514	4.2%	~27,800 edges
DrugBank	1,706	191,808	13.2%	—

Methodological improvement: the DDI-LLM notebook filters out drug pairs where either drug lacks a valid SMILES string *or* a valid description, reducing BioSnap from 48,514 to ~27,800 edges. This filter is unnecessarily applied even for description-only embeddings. In this project, the filter is applied conditionally: only SMILES-based embeddings require valid SMILES; description-based embeddings use the full graph (~47,000 edges), recovering a 69% increase in training data.

OGB-ddi

OGB-ddi is a standardised DDI benchmark from the Open Graph Benchmark (Stanford) [9], enabling direct comparison with published methods including LLM-DDI. It contains FDA-approved and experimental drugs connected by known interactions, using a protein-target split that ensures the test set contains drugs targeting different proteins than training drugs.

	Drugs	Interactions	Sparsity	Mean degree	Components	Node features
BioSnap	1,514	48,514	4.2%	64.1	2	via desc/SMILES
DrugBank	1,706	191,808	13.2%	224.4	—	via desc/SMILES
OGB-ddi	4,267	1,067,911	11.7%	500.5	1	None

OGB-ddi has no pre-computed node features (`graph.x = None`). This means LLM embeddings must be generated from scratch using the DrugBank ID mapping provided by OGB. The graph is perfectly connected (single component, no isolated nodes), and its density is substantially higher than BioSnap, which introduces more noise during GCN training but also provides richer neighbourhood information per node.

Split: train 1,067,911 / val 133,489 / test 133,489 edges.

Negative samples — CRESCENDDI

The original DDI-LLM pipeline samples negatives randomly from unlabelled drug pairs — methodologically weak since unlabelled pairs may be undiscovered positives. **CRESCENDDI** [8] provides 4,544 clinically verified non-interacting pairs. A mapping to DrugBank IDs was performed using the DrugBank vocabulary file (55,576 name-ID entries covering 19,853 drugs):

	BioSnap	DrugBank
Unique CRESCENDDI drugs	434	
Mapped successfully	4,535 / 4,544 (99.8%)	
Usable negative pairs	4,306 (95.0%)	4,078 (89.9%)

The 9 unmapped pairs correspond to the truncated name *mefenamate*. Class imbalance (ratio $\approx 1:6$ for BioSnap) is handled via weighted BCE loss: `pos_weight = n_pos / n_neg`.

Data Analysis

BioSnap network structure

Property	Value	Interpretation
Largest connected component	1,510 / 1,514	4 peripheral drugs
Average degree	64.1	moderate connectivity
Degree assortativity	negative	hubs connect to low-degree nodes
Mean clustering coefficient	~ 0.25	bimodal distribution
Mean closeness centrality	0.40–0.45	2–3 hops between any two drugs
Degree vs. desc. length corr.	0.14	weak — embeddings add new info

The degree distribution is heavy-tailed: 38% of drugs have fewer than 15 interactions while a small number of hub drugs interact with over 400 others. Betweenness centrality is concentrated near zero (90% of drugs), indicating structural robustness. Closeness centrality follows a near-normal distribution centred around 0.40–0.45, confirming that most drugs are reachable within 2–3 hops — consistent with the 3-layer GCN architecture. The weak correlation ($r=0.14$) between degree and description length confirms that graph structure and LLM embeddings capture complementary information.

OGB-ddi network structure

Property	Value	Interpretation
Connected components	1	fully connected
Mean degree	500.5	8 $\times$ higher than BioSnap
Median degree	446.0	most drugs well-connected
Max degree	2,234	extreme hub drug
Min degree	1	no isolated nodes
Sparsity	11.7%	denser than BioSnap
Node features	None	must generate embeddings

OGB-ddi is substantially denser than BioSnap. The average drug interacts with 500 others, compared to 64 in BioSnap. This density introduces more noise during GCN message passing, which is consistent with the LLM-DDI paper finding OGB-ddi harder to model despite its denser structure. Since node features are absent, generating biomedical LLM embeddings (PubMedBERT) from DrugBank descriptions is not merely an improvement but a prerequisite for the method to work at all on this dataset.

Description coverage: all 4,267 OGB-ddi drugs have entries in the DrugBank description file (100% coverage), enabling PubMedBERT embedding generation for all nodes without fallback. However, 662 drugs (15.5%) have corrupted descriptions (;;; Excel artefacts), substantially higher than BioSnap (0.5%). These drugs will receive zero-vector embeddings unless a SMILES-based fallback (ChemBERTa) is applied.

Clustering coefficient: the approximate mean clustering coefficient is 0.516, approximately twice that of BioSnap (~ 0.25), indicating that OGB-ddi has much denser local community structure. Drugs that share interaction partners are highly likely to interact with each other, which should support consistent GCN neighbourhood aggregation.

Degree vs description length: the correlation between node degree and description length is weak ($r=0.174$), consistent with BioSnap ($r=0.14$), confirming that graph structure and text embeddings capture complementary information across both datasets.

Description quality

Feature	BioSnap		DrugBank	
	Count	%	Count	%
Valid descriptions	1,489/1,514	98.3%	1,695/1,706	99.4%
Corrupted	8/1,514	0.5%	6/1,706	0.4%
Missing	17/1,514	1.1%	5/1,706	0.3%
Valid SMILES	1,332/1,514	88.0%	1,695/1,706	99.4%

Two corruption types: ; ; ; (n=6,512) and #NAME? (n=6), both Excel artefacts. Only 14 drugs across both datasets are affected. Median description length is 90 characters (std=245). The weak correlation with degree (r=0.14) means low-degree drugs may have either short or long descriptions independently — an important limitation for LLM embedding quality.

Implementation

Pipeline status

The DDI-LLM pipeline has been successfully reproduced on BioSnap using Google Colab Pro (H100 GPU). A bug was identified and fixed in the LM() embedding loading function: the original code dropped only 2 metadata columns for all embedding files, but the BERT description file contains 3 metadata columns (DrugBank ID, SMILES, and an index), causing SMILES strings to be incorrectly used as embedding features. The fix identifies metadata columns dynamically by dtype:

```
metadata_cols = [col for col in df.columns if df[col].dtype == object]
features = df.drop(columns=metadata_cols)
```

First results on BioSnap

Embedding	AUROC	AUPR
GPTDesc	0.974	0.934
BERTDesc	0.558	0.458

GPTDesc achieves 0.974 AUROC, consistent with published results. BERTDesc performs near-random (0.558) — consistent with the original paper’s results which also show BERTDesc underperforming across all learning rates (max 0.750). This motivates PubMedBERT as a domain-specific replacement. Results are from a single seed at lr=0.0001; multi-seed evaluation is pending.

Embeddings

Embedding	Input	Type	Dim	Status
BERT	Description	General LLM	768	Available (bug fixed)
GPT	Description	General LLM	1,536	Available, working
LLaMA/LLaMA2	Description	General LLM	—	Available
ChemBERTa	SMILES	Chemistry LLM	768	Available (newer notebook)
Molformer	SMILES	Chemistry LLM	—	Available (newer notebook)
PubMedBERT	Description	Biomedical LLM	768	To generate
Morgan FP	SMILES	Classical	1,024	To generate (RDKit)

Environment

Google Colab Pro (H100 GPU) for model training and embedding generation. Local VS Code (Windows 11, Python 3.12, `.venv`) for data exploration and scripting. Packages are cached to Google Drive to avoid reinstallation overhead. Key versions: `torch==2.11.0+cu130`, `torch_geometric==2.7.0`, `transformers==5.5.4`, `ogb==1.3.6`, `numpy==1.26.4`, `pandas==3.0.1`, `rdkit`, `scikit-learn`.

Next Steps

Task	Details
Complete OGB-ddi analysis	Clustering coefficient, description coverage, degree vs description correlation
Fix SMILES filter	Apply filter conditionally per embedding type; recover $\sim 47,000$ edges for description embeddings
Integrate CRESCENDDI negatives	Replace structured negative sampling with 4,306 confirmed negatives + weighted BCE loss
Generate PubMedBERT embeddings	<code>microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract</code> ; run once on Colab H100, save to Drive
Run 5 seeds per configuration	Wrap training loop in seed loop; report mean $\pm$ std AUROC and AUPR
Classical baselines	Logistic regression, random forest, MLP on concatenated drug pair embeddings

References

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